WhatNext Vision Motors

Salesforce CRM - Vehicle Order Management System

FUNCTIONAL SPECIFICATION DOCUMENT (FSD)

Project Demonstration Phase

1. Document Overview

Item	Details
Project	WhatNext Vision Motors - Vehicle Order Management System
Platform	Salesforce Developer Edition (Lightning)
Document type	Functional Specification Document
Version	1.0

1.1 Purpose

This document describes what the WhatNext Vision Motors CRM must do: its data, user interface, automation and business rules. It is the reference for development and review.

1.2 Scope

In scope: management of vehicles, dealers, customers, orders, test drives and service requests; automatic dealer assignment; test-drive reminder email; stock validation and update; scheduled processing of pending orders. Out of scope: customer-facing portals, payment processing and external integrations.

2. Users and Access

User	Description	Access
System Administrator	Configures and uses the WhatNext Vision Motors app	Full access to all objects, tabs, reports and dashboards
Customer (record)	Person who orders, test drives or requests service	Represented as Vehicle_Customer__c records; receives email reminders

3. Functional Requirements

ID	Requirement	Delivered by
FR-01	The system shall store vehicle, dealer, customer, order, test-drive and service-request records.	Custom Objects
FR-02	Each object shall be reachable from a custom tab.	Custom Tabs
FR-03	All objects, reports and dashboard shall be available in one Lightning app named WhatNext Vision Motors.	App Manager
FR-04	A vehicle shall reference its dealer; orders, test drives and service requests shall reference customer and vehicle.	Lookup Relationships
FR-05	When a new order is created with status Pending, the system shall assign the nearest dealer based on customer address or location.	Auto Assign Dealer Flow
FR-06	When a test drive has status Scheduled, the system shall email the customer one day before Test_Drive_Date__c.	Test Drive Reminder Flow

ID	Requirement	Delivered by
FR-07	The system shall block an order when the vehicle's Stock_Quantity__c is zero or less.	Apex Trigger (before)
FR-08	When an order has status Confirmed, the system shall reduce the vehicle stock by one.	Apex Trigger (after)
FR-09	The system shall process Pending orders in batches, confirming eligible orders when stock is available and reducing stock.	Batch Apex
FR-10	The batch process shall run automatically on a schedule using a batch size of 50.	Scheduled Apex

4. Data Specification

4.1 Objects

Object	Purpose	Relationships
Vehicle__c	Stores vehicle details	Dealer and Orders
Vehicle_Dealer__c	Stores authorized dealer information	Orders
Vehicle_Customer__c	Stores customer details	Orders and Test Drives
Vehicle_Order__c	Tracks vehicle purchases	Customer and Vehicle
Vehicle_Test_Drive__c	Tracks test-drive bookings	Customer and Vehicle
Vehicle_Service_Request__c	Tracks vehicle servicing requests	Customer and Vehicle

4.2 Fields

Object	Key Fields
Vehicle__c	Vehicle_Name__c (Text); Vehicle_Model__c (Picklist); Stock_Quantity__c (Number); Price__c (Currency); Dealer__c (Lookup); Status__c (Picklist)
Vehicle_Dealer__c	Dealer_Name__c (Text); Dealer_Location__c (Text); Dealer_Code__c (Auto Number); Phone__c (Phone); Email__c (Email)
Vehicle_Customer__c	Customer_Name__c (Text); Email__c (Email); Phone__c (Phone); Address__c (Text); Preferred_Vehicle_Type__c (Picklist)
Vehicle_Order__c	Customer__c (Lookup); Vehicle__c (Lookup); Order_Date__c (Date); Status__c (Picklist)
Vehicle_Test_Drive__c	Customer__c (Lookup); Vehicle__c (Lookup); Test_Drive_Date__c (Date); Status__c (Picklist)
Vehicle_Service_Request__c	Customer__c (Lookup); Vehicle__c (Lookup); Service_Date__c (Date); Issue_Description__c (Text); Status__c (Picklist)

4.3 Relationships

- Vehicle__c → Vehicle_Dealer__c (Dealer__c lookup)
- Vehicle_Order__c → Vehicle_Customer__c (Customer__c) and Vehicle__c (Vehicle__c)

- Vehicle_Test_Drive__c → Vehicle_Customer__c and Vehicle__c
- Vehicle_Service_Request__c → Vehicle_Customer__c and Vehicle__c

5. User Interface Specification

The Lightning app **WhatNext Vision Motors** shall contain these navigation items: Vehicle, Dealer, Customer, Order, Test Drive, Service Request, Reports and Dashboard. The System Administrator profile is assigned to the app. Each object uses a custom tab with the default profile and app visibility settings.

6. Automation Specification

6.1 Auto Assign Dealer (Record-Triggered Flow)

Attribute	Specification
Object	Vehicle Order
Trigger	A record is created
Entry condition	Status__c Equals Pending
Steps	Get Records (customer by customer ID) → Get Records (nearest dealer by customer address/location) → Update Records (assign dealer to order)
Flow name	Auto Assign Dealer (activated)

6.2 Test Drive Reminder (Record-Triggered Flow)

Attribute	Specification
Object	Vehicle Test Drive
Trigger	A record is created or updated
Entry condition	Status__c Equals Scheduled
Scheduled path	One day before Test_Drive_Date__c
Steps	Get customer information → Send Email
Email subject	Reminder: Your Test Drive is Tomorrow!
Recipient	Customer Email__c

7. Business Rules (Apex)

Rule	Context	Behaviour
BR-01 Stock validation	before insert / before update	If Stock_Quantity__c <= 0, prevent the order with the message: *This vehicle is out of stock. Order cannot be placed.*
BR-02 Stock update	after insert / after update	If Status__c = Confirmed, reduce the vehicle Stock_Quantity__c by one and save the vehicle.

Rule	Context	Behaviour
BR-03 Batch confirmation	VehicleOrderBatch	For Pending orders with stock available: set Confirmed, reduce stock by one, save.
BR-04 Scheduling	VehicleOrderBatchScheduler	Launch VehicleOrderBatch with batch size 50 at the scheduled interval.

Trigger handler structure

```
public class VehicleOrderTriggerHandler {
  public static void handleTrigger(List<Vehicle_Order__c> newOrders,
      Map<Id, Vehicle_Order__c> oldOrders, Boolean isBefore,
      Boolean isAfter, Boolean isInsert, Boolean isUpdate) {
    if (isBefore && (isInsert || isUpdate)) {
      preventOrderIfOutOfStock(newOrders);
    }
    if (isAfter && (isInsert || isUpdate)) {
      updateStockOnOrderPlacement(newOrders);
    }
  }
}
```

8. Process Flow

Order lifecycle: New Pending Order → Auto Assign Dealer → Stock validated (Apex before) → Order confirmed (manually or by batch) → Stock reduced by one.

Test drive lifecycle: Test drive Scheduled → one day before date → reminder email sent to customer.

9. Acceptance Criteria

ID	Scenario	Expected result
AC-01	Create a Pending order for a vehicle	Dealer is populated on the order automatically
AC-02	Create an order for a vehicle with stock 0	Order is rejected with the out-of-stock message
AC-03	Confirm an order for a vehicle in stock	Vehicle stock decreases by one
AC-04	Schedule a test drive	Customer receives the reminder email one day before the date
AC-05	Run VehicleOrderBatch with Pending orders and available stock	Eligible orders become Confirmed and stock is reduced
AC-06	Wait for the scheduled interval	Batch job starts automatically and is visible in Setup

10. Assumptions and Constraints

- The solution runs on a Salesforce Developer Edition org.
- Dealer assignment depends on customer address or location data being filled in.

- Reminder emails depend on a valid customer Email__c value.
- Each confirmed order reduces stock by exactly one unit.